PAVAN PONUGOTI

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SUMMARY

Data Engineer with 5 years of experience designing and operating ETL/ELT pipelines, cloud data platforms, and real-time streaming systems using Python, SQL, Spark, Kafka, and Databricks, with deep expertise in AWS, Azure, and Snowflake. Proven track record owning data workflows end-to-end across ingestion, transformation, orchestration, and reporting; building reusable DBT models, Airflow DAGs, and shared data-quality frameworks that internal teams rely on; and modernizing legacy on-prem systems into cloud-native architectures on AWS and Azure. Strong fundamentals in distributed data processing, query optimization, data governance, and root-cause analysis for pipeline reliability; experienced partnering with cross-functional analytics, product, and business teams to translate raw data into actionable, cost-efficient insights. Holds a Master of Science in Computer and Information Sciences from New England College.

SKILLS

Languages / Databases: Python (Pandas, NumPy, Scikit-learn), R, SQL, MySQL, Oracle SQL, DynamoDB

ETL & Data Integration: IBM DataStage, DBT, Snowflake (Snowpipe, Streams & Tasks, Time Travel), SAS, SSIS, Azure Data Factory (ADF), AWS Glue, Airflow

Cloud: AWS (Lambda, S3, DynamoDB, EMR, Step Functions, Redshift, Glue), Azure (ADF, ADLS, Synapse, Cosmos DB), Terraform

Big Data / Streaming: Spark, PySpark, Kafka, Hadoop, Hive, Kudu, Databricks (Delta Lake, Unity Catalog, MLflow), Kubernetes

Data Warehousing & Modeling: Dimensional Modeling (Star/Snowflake Schema), Data Lake & Lakehouse Architecture, Medallion Architecture, CDC, Data Governance, MDM

Testing & Data Quality: Pytest, dbt tests, Great Expectations, SQLFluff, Data Quality Frameworks

Visualization & Tools: Tableau, Power BI, Git, Jenkins, Jira, Confluence, Agile, CI/CD, Jupyter Notebooks

WORK EXPERIENCE

Senior Data Engineer — **May 2025–Present**

Capital One — *Richmond, VA*

- Architected scalable ETL/ELT pipelines across multiple source systems into centralized cloud data warehouses, boosting data availability for reporting by 25% and cutting manual intervention.
- Leveraged Snowpark to build in-warehouse Python data transformations, eliminating the need to move data to external compute clusters for feature engineering and scoring workloads.
- Implemented Delta Live Tables on Databricks to declaratively orchestrate multi-stage Bronze/Silver/Gold pipelines, using built-in data quality expectations and automatic dependency resolution to simplify maintenance.
- Utilized Unity Catalog to centralize data governance, column-level masking, and lineage tracking across Databricks workspaces, strengthening audit readiness and cross-team data discoverability.
- Optimized complex SQL queries and data models for high-volume reporting workloads, reducing average query runtime by 30% and accelerating business intelligence delivery.
- Built automated data quality and monitoring frameworks with real-time alerting, cutting production data incidents by 35% and improving pipeline reliability SLAs.
- Designed incremental data ingestion frameworks using AWS Glue and Lambda, handling schema drift and late-arriving records across multiple upstream source systems.
- Implemented CI/CD-driven deployment workflows for data pipelines using Git and Jenkins, reducing release cycle time by 40% and minimizing production defects.
- Migrated legacy batch reporting workflows to event-driven Kafka-based architectures, replacing scheduled batch jobs with continuous stream processing for downstream consumers.
- Built parameterized DBT and Airflow frameworks enabling self-service pipeline onboarding for new data sources, cutting new-pipeline setup time by 45%.

Data Engineer — **Jan 2024–May 2025**

Discover — *Houston, TX*

- Drove data analysis workflows end-to-end using Python and SQL across large-scale operational datasets; engineered AWS Lambda functions triggered by S3 events to stream data into DynamoDB for low-latency downstream access.

- Re-architected ETL/ELT pipelines with DBT and SQL-based transformations, and tuned Snowflake workloads via clustering, caching, and query optimization to slash query latency.
- Automated serverless workflows with AWS Step Functions to orchestrate multi-stage Glue jobs, coordinating dependency handling and error retries before publishing curated datasets to S3 and Redshift.
- Delivered PySpark/Spark SQL transformations on AWS EMR and authored Airflow DAGs to orchestrate ETL/ELT workflows; built Tableau dashboards that sharpened executive visibility into KPIs.
- Established data lineage, governance, and quality frameworks in partnership with cross-functional teams, strengthening reliability of enterprise reporting pipelines.
- Implemented root-cause analysis procedures to diagnose and resolve ETL job failures, improving overall system reliability and reducing downtime by 18%.
- Troubleshot production issues using CloudWatch logs and metrics, cutting mean time-to-resolution for pipeline incidents by 25%.

Data Engineer **May 2021–Jul 2023**
Accenture *Hyderabad, India*

- Engineered and automated ETL workflows using IBM DataStage and Azure Data Factory, unifying data across SQL Server, MongoDB, and cloud platforms for enterprise-wide reporting.
- Migrated on-premises datasets into Azure Data Lake Storage via ADF (V1/V2) pipelines and built real-time streaming apps with Spark Streaming and Kafka into Apache Kudu on Cloudera Hadoop.
- Built reusable DBT models and integrated Snowflake Cloud Data Warehouse into the architecture, powering centralized, high-performance analytics for stakeholders.
- Automated data cleansing and analysis processes, boosting accuracy and slashing processing time by 40%; delivered Tableau dashboards that drove faster business decisions.
- Designed and implemented ETL pipelines integrating Azure Cosmos DB into Azure Synapse Analytics, leveraging Azure Functions for orchestration and transformation.
- Leveraged SAS for transformation and validation during statistical reporting and migration testing, ensuring data accuracy across legacy-to-cloud migrations.

EDUCATION

Master of Science, Computer and Information Sciences
New England College, Henniker, NH — Aug 2023–May 2025

PROJECTS

- Dry Beans Classification (SVM, Random Forest, Deep Learning, KNN) — Built and evaluated multiple classification models on a structured agricultural dataset, applying data cleaning and preprocessing techniques to improve data quality and model accuracy.
- Goodreads Data Visualization (Data Cleaning, Transformation, Visualization) — Processed and refined a large-scale dataset, applying data cleaning and transformation techniques to improve data quality for downstream analysis and reporting.